KALSHI VS. FED FUNDS FUTURES

Prediction Markets as Macro Data

Are Kalshi's prices on FOMC decisions good forecasts, and how do they compare with fed funds futures?

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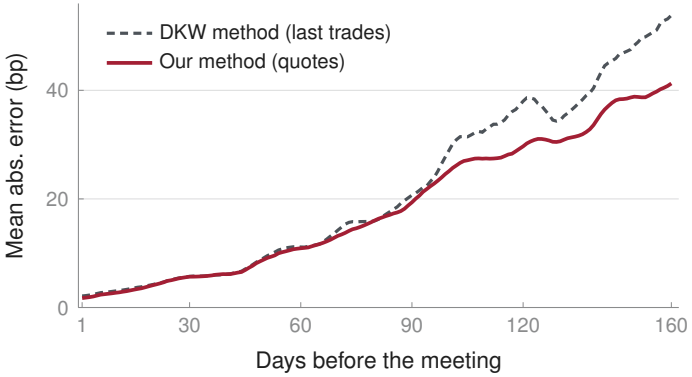
Progress update | October 2026

MIT Sloan School of Management

15.S06 AI and ML Research in Finance

We replicate Diercks–Katz–Wright and improve on it

Kalshi-implied policy rate vs. the rate actually set, 41 FOMC meetings 2021–2026



41/41

meetings called correctly the day before

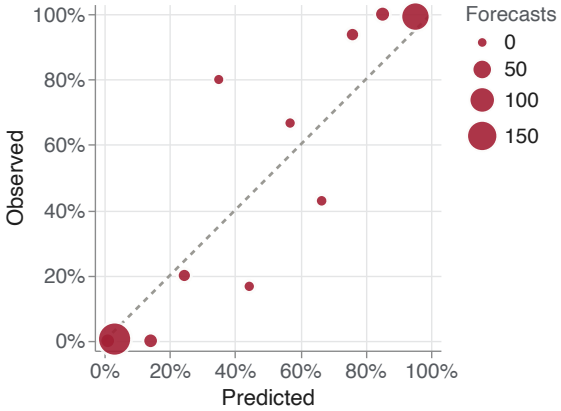
−14%

average error over 1–160 days (17.9 vs. 20.8bp)

Why: 40–45% of last trades are over a week old.

Fed-decision prices show a favorite–longshot pattern

Share of contracts that paid out, by price, 1–21 days before the meeting



0.5%

payout rate of contracts priced 1–10¢ (avg. price 3.1¢)

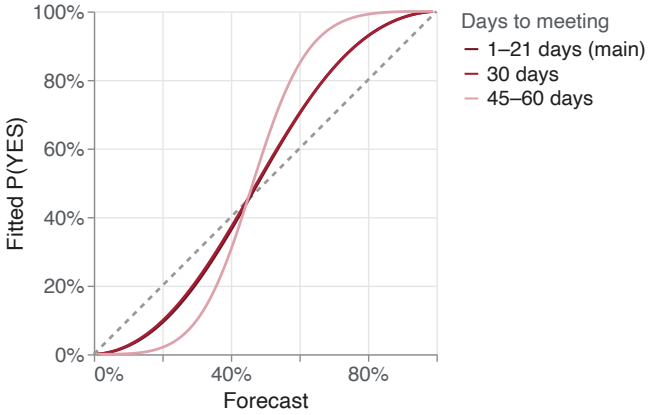
99.3%

payout rate of contracts priced 90–99¢ (avg. 95.3¢)

Bubble size = forecasts. 632 forecasts, 28 meetings. Middle bins hold 2–6 meetings each.

A calibration regression confirms the pattern

$P(\text{YES}) = \Lambda(\alpha + \beta \text{ logit } p)$; calibrated means $\alpha = 0$, $\beta = 1$



1.75

calibration slope β at 1–21 days (SE 0.45; bootstrap 95% CI 1.22–3.54)

p = 0.01

joint test of calibration (cluster bootstrap)

$\beta > 1$: favorites win more than priced, longshots less. Errors clustered by meeting (28). No significant change after the May 2025 liquidity jump.

Futures give a benchmark probability for every meeting

The WIRP-style method: from futures prices to outcome probabilities

Futures price
FF1–FF13, daily

→

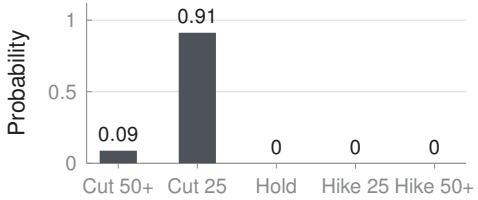
Avg. rate per month
$100 - P$

→

Rate after meeting
un-blend the meeting month

→

Expected move x
in 25bp steps

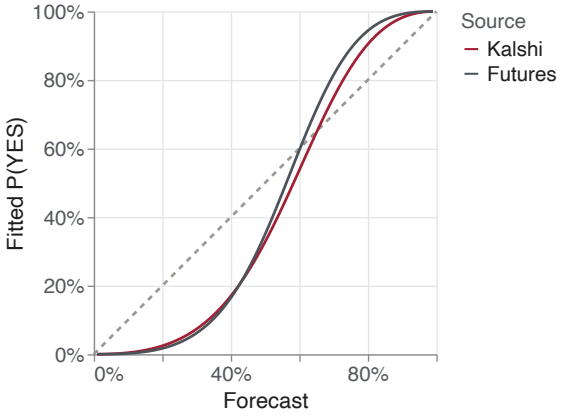


Example: 30 Jul 2024, for the 18 Sep meeting.
$x = -1.09$ steps $\Rightarrow$ 91% cut 25bp, 9% cut 50bp.

All mass sits on the two outcomes next to x, so the futures can never price the tails.

Kalshi is calibrated like futures and slightly more accurate

Hold and ±25bp outcomes, both read at 15:00 ET and scored against the decision



2.14 vs. 2.48

calibration slope β, Kalshi vs. futures (1–21 days): both under-confident

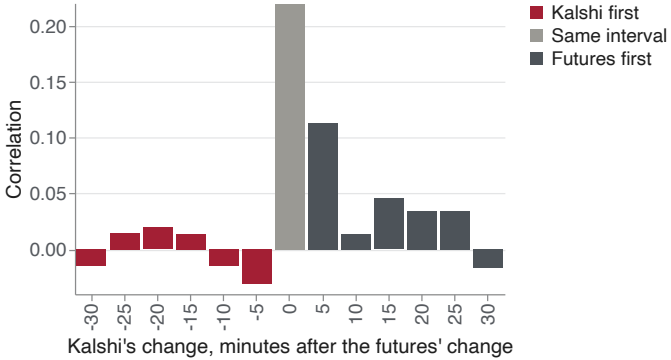
0.039 vs. 0.046

Brier score at 30 days, Kalshi vs. futures

Kalshi scores better at 1–30 days, futures at 45–60 days (few meetings). Kalshi prices moves beyond 25bp higher.

Intraday, futures lead Kalshi by about five minutes

Correlation of 5-minute changes in the expected move, Mar–Oct 2026, 5 meetings



0.22

same 5 minutes: most co-movement is simultaneous

$p < 0.001$

past futures moves predict Kalshi (Granger; p = 0.06 the other way)

16%

of a large futures move shows up in Kalshi within 15 minutes

Appendix

Research questions and next steps

Questions

- **RQ1 Calibration.** Are Kalshi prices rational forecasts? Favorite–longshot bias? Better than futures and surveys? *First results in.*
- **RQ2 Price discovery.** How fast do prices absorb CPI, payrolls and FOMC news vs. futures? *Lead–lag started.*
- **RQ3 Forecasting.** Use Kalshi distributions to predict CPI, unemployment and GDP. *Not started.*

Next steps

- Lead–lag around FOMC statements (14:00 ET) vs. quiet times; extend the intraday futures sample before March 2026.
- Information shares (Hasbrouck) next to the Granger tests.
- Formal Kalshi vs. futures score test (e.g. Diebold–Mariano).
- Rerun the daily futures comparison on refreshed candles (meetings after Sep 2025).
- Add SPF and Bloomberg consensus benchmarks for CPI and payrolls.

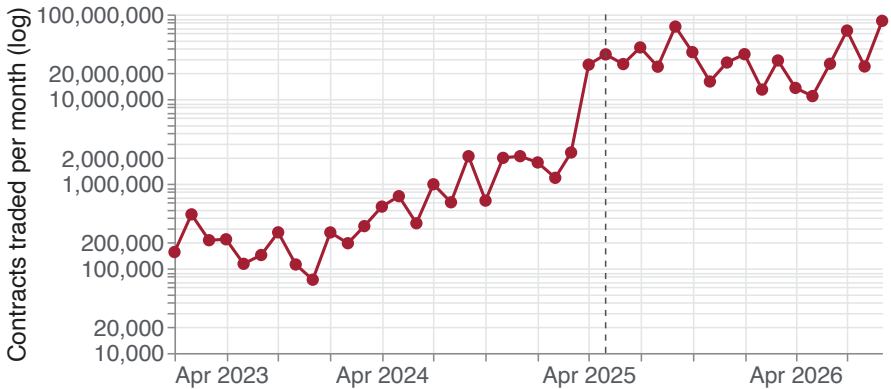
Dataset: every Kalshi trade and 1-minute quote

Series	Events	Contracts	Trades	Volume	Since
Fed decision (KXFEDDECISION)	40	200	770,571	620M	Apr 2023
CPI m/m (KXCPI)	67	552	204,598	25.5M	Jul 2021
Core CPI m/m (KXCPICORE)	56	500	33,259	7.7M	Jul 2022
CPI y/y (KXCPIYOY)	49	713	241,413	37.9M	Dec 2022
Payrolls (KXPAYROLLS)	45	422	116,507	19.6M	Mar 2023
Fed funds ladders (KXFED)	52	660	$\sim$222,000	59M	Jul 2021

- 10.4M one-minute candles. 100% of 2,155 settled contracts match Kalshi's reported volume within $\pm$1%.
- Futures: Bloomberg FF1–FF13, EFFR and target range, daily 2000–2026; 1-minute bid/ask bars Mar–Oct 2026.
- Fed outcomes: 19 holds, 5 cuts of 25bp, 1 cut of 50bp (Sep 2024), 3 hikes of 25bp.

Kalshi has become a deep market for Fed decisions

Contracts traded per month, all KXFEDDECISION meetings (log scale)



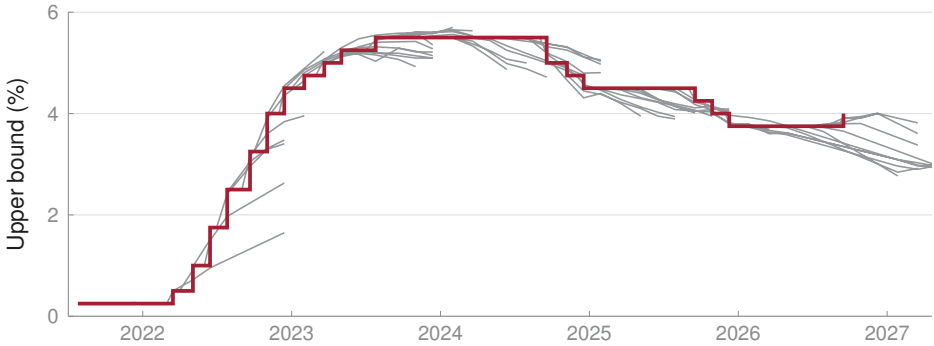
620M contracts since 2023; volume rose about 10× in spring 2025. Median bid–ask spread 1¢.

Literature: what is known and what we add

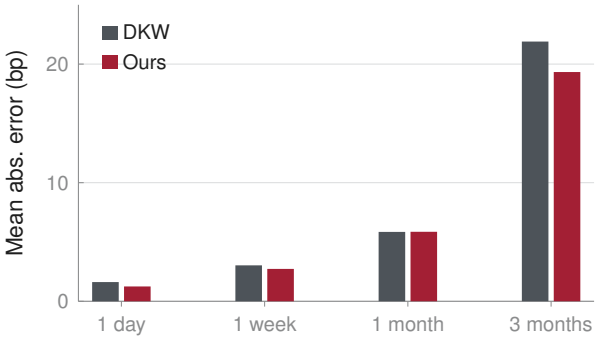
Paper	Finding	Gap we target
Diercks, Katz & Wright (2026)	Kalshi distributions for fed funds, CPI, GDP; CPI beats Bloomberg consensus	No Brier/log scores, no bias test, daily only
Bürgi, Deng & Whelan (2026)	Strong favorite–longshot bias overall; weak in economics contracts	Test on macro with clean specification
Bartlett & O'Hara (2026)	Microstructure of 41.6M Kalshi trades	No forecast evaluation
Wolfers & Zitzewitz (2004)	Contract design reveals mean, median, distribution	Early evidence: 16 obs.
Kelly et al. (2026)	Transformer stochastic discount factor	Only regularized ideas fit our sample (RQ3)

Kalshi-implied rate paths vs. the realized target

Each line: the path the market implied on one date; red steps: the actual upper bound



Replication accuracy by horizon



vs. NY Fed dealer survey

132 forecasts, 37 meetings: survey 19.1bp, Kalshi mode 17.0bp.

Method

Mid quote if spread $\leq$ 20¢, else last trade (20–40¢) or dropped; weighted isotonic fit across strikes.

On DKW's 2022–25 window

19.9 vs. 22.7bp (1–160 days).

Calibration regression: estimates

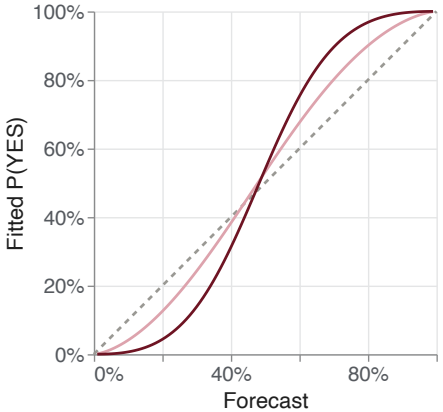
Logit of outcome on logit of price; meeting-clustered SEs; 999-draw cluster bootstrap

Horizon	Obs. / meetings	α (SE)	β (SE)	β 95% CI (boot)	p(β=1)	p joint (boot)
1–21 days (main)	454 / 28	0.141 (0.030)	1.751 (0.453)	[1.22, 3.54]	0.097	0.010
30 days	78 / 27	0.165 (0.062)	1.723 (0.533)	[1.11, 2.77]	0.175	0.008
45–60 days	100 / 18	0.445 (0.186)	3.133 (1.058)	[1.97, 13.66]	0.044	0.189

- $\beta > 1$ at every horizon: under-confident prices, the favorite–longshot pattern.
- $\alpha > 0$ is significant: the YES side of the normalized buckets pays slightly more than priced.
- Sample: normalized prices at 16:00 ET, tick-floor contracts excluded; 632 of 1,384 forecasts.

No detectable change after the liquidity jump

Fitted calibration curves, 1–21 days: 16 meetings before May 2025, 12 from May 2025



1.49 → 2.35

β before vs. from May 2025

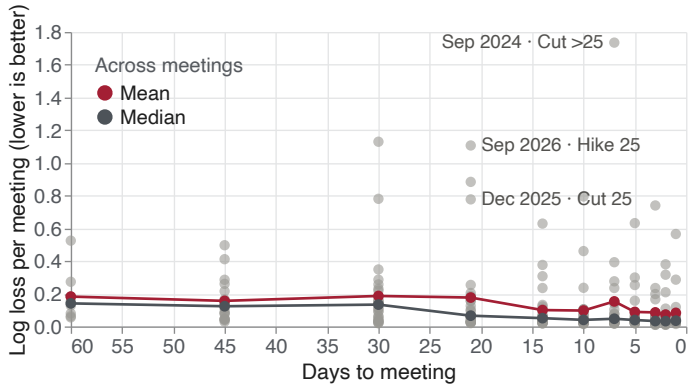
p = 0.19

difference in β (joint test p = 0.36, bootstrap)

A large p-value means no detectable change, not no change. The regimes also differ in policy: mostly holds before, cuts after.

Accuracy fades with horizon, and single surprises dominate

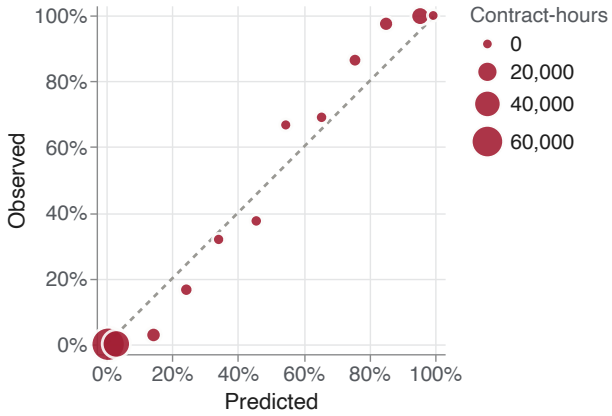
Log loss per meeting by days to meeting; labelled points are the largest misses



Mean log score 0.07 at 1 day, 0.28 at 30 days; median at 1–21 days 0.05 vs. mean 0.14.

Minute-level reliability shows the same S-shape

Descriptive only: raw mid quotes every minute, 0–60 days, includes meeting day



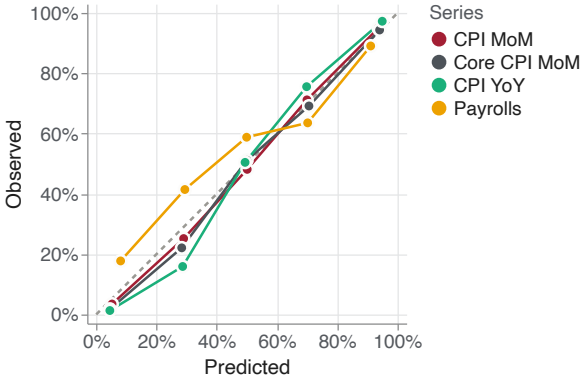
Specification fixed before any significance test

- Forecast: 16:00 ET bid–ask mid, $h \in \{1, 2, 3, 5, 7, 10, 14, 21, 30, 45, 60\}$ days before.
- Horizon groups: 1–21 days (main), 30, 45–60.
- Fed buckets normalized to sum to 1 (overround $\approx 1.5\%$); raw as robustness.
- Tick-floor quotes (below 1¢, 0¢ bid) excluded; included as robustness.
- Stale quotes kept and flagged; drop-stale as robustness. No spread filter.
- Scores per contract (log loss, calibration regression) and per meeting (log score, Brier).
- Minute-level results are descriptive only.
- Voided contracts and one untraded duplicate Fed event dropped.

With about 28 meetings the test has low power, so the specification is fixed in advance.

CPI is under-confident, payrolls over-confident

1–21 days before the release; 18,049 forecasts, 203 releases; SEs clustered by release date



Series	β (SE)	p(β=1)
CPI m/m	1.20 (0.10)	0.043
Core CPI m/m	1.25 (0.11)	0.025
CPI y/y	1.40 (0.19)	0.033
Payrolls	0.75 (0.10)	0.015
CPI pooled	1.28 (0.10)	0.005
All series	1.11 (0.07)	0.123

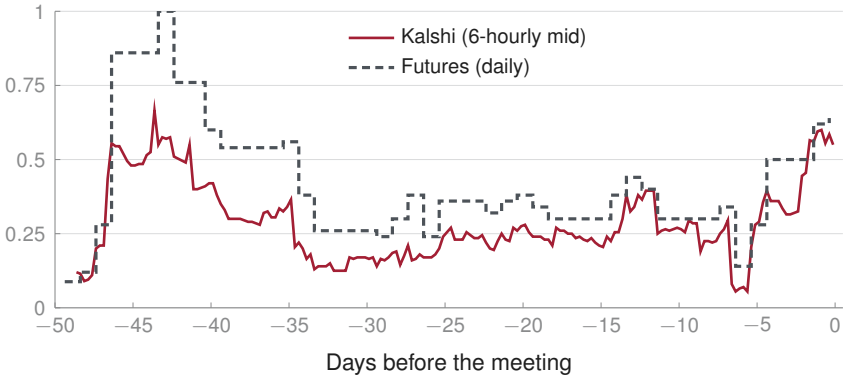
Reliability by series (left). Every α is insignificant. Ladder contracts are not exclusive, so prices are not normalized.

How futures-implied probabilities are computed

- FFn settles on the average EFFR of month $n-1$, so the implied average for month j is $\bar{r}_j = 100 - P_{j+1}$.
- A meeting on day d of an N-day month splits it: $\bar{r}_j = \frac{d}{N} r_{\text{pre}} + \frac{N-d}{N} r_{\text{post}}$.
  - Method A solves for r_{post} and amplifies noise by $N/(N-d)$.
  - Method B (preferred) takes r_{post} from the next meeting-free month.
- Meetings are chained; the start rate is the range midpoint plus a 10-day median EFFR spread.
- Expected move x in 25bp steps: with $n = \lfloor x \rfloor$, $p = x - n$, the move is n steps w.p. $1-p$ and $n+1$ steps w.p. p.
- Checks: all 2,909 (meeting, date) pairs sum to 1; Methods A and B differ by a median of 1.4bp.

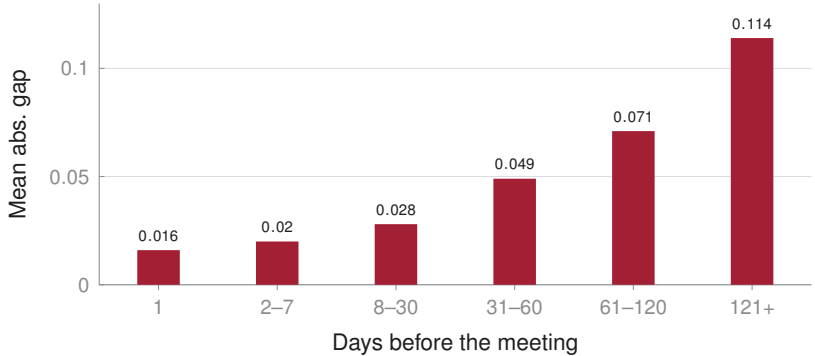
One meeting up close: September 2024

Probability of a cut larger than 25bp; the Fed cut 50bp



Kalshi and futures agree closely near the meeting

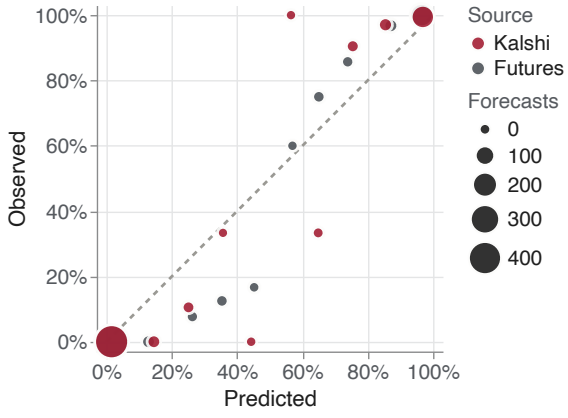
Mean absolute gap in probability, hold and ±25bp outcomes; 14,513 contract-days



Correlation falls from 0.998 (1 day) to 0.928 (121+ days).

Reliability: futures vs. Kalshi

Hold and ±25bp outcomes, 1–30 days pooled; bins from 0.3 to 0.6 hold 2–5 meetings



Scores: Kalshi vs. futures by horizon

Days before	1	7	21	30	45	60
Brier, futures	0.0049	0.0198	0.0425	0.0459	**0.0222**	**0.0532**
Brier, Kalshi	**0.0044**	**0.0191**	**0.0383**	**0.0392**	0.0310	0.0755
Log loss, futures	0.035	**0.081**	0.148	0.170	**0.113**	**0.192**
Log loss, Kalshi	**0.030**	0.084	**0.138**	**0.153**	0.142	0.253

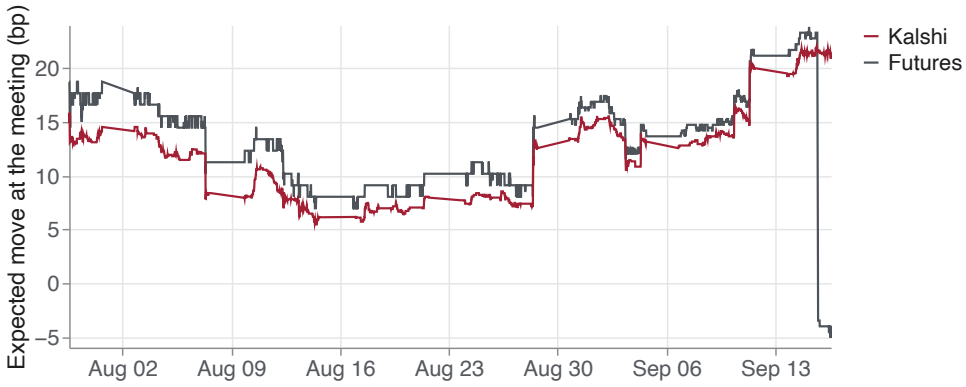
Bold: lower (better) score. 28 meetings at 1–21 days, 8 at 60 days. Probabilities clipped to [0.005, 0.995].

Lead–lag: data and method

- **Futures:** Bloomberg 1-minute bid/ask bars, 30 Mar – 12 Oct 2026, contracts Mar–Dec 2026; expected move per minute with the daily method (A or B per meeting). Median spread 0.5bp.
- **Kalshi:** expected move $\sum_k q_k \cdot \text{move}_k$ from 1-minute bucket mids, rescaled to sum to 1, carried onto the CME minute grid.
- **Alignment:** Bloomberg bars are labelled by start minute, so they are shifted one minute to match Kalshi. Without it the futures appear to lead by exactly one minute.
- **Validation:** at 15:00 the minute series matches daily settlements: 135 days, correlation 0.998, median gap 0.27bp.
- **Tests** on 5-minute changes (95% of futures and 92% of Kalshi changes are zero): cross-correlations, Granger regressions with 6 lags clustered by session, and paths after the largest 0.1% of 1-minute moves.

The two markets track the same expectations

Expected move at the 16 Sep 2026 meeting, every 15 minutes (CME hours)



Granger tests: futures predict Kalshi, much less the reverse

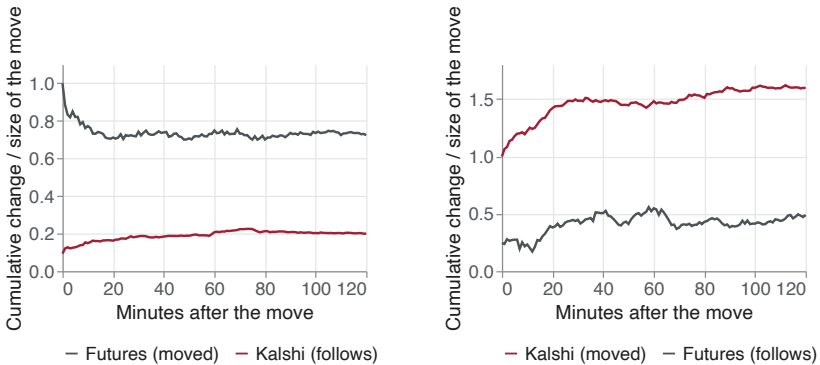
Question	Wald χ^2 (6 df)	p-value	Sum of cross lags (SE)
Do past futures moves predict Kalshi?	30.30	$<$0.001	0.133 (0.030)
Do past Kalshi moves predict futures?	12.02	0.062	0.279 (0.090)

37,632 five-minute changes, 140 sessions (all CME hours), 6 lags = 30 minutes, SEs clustered by session.

Caveats: about 140 days and five meetings; mostly-zero changes; different tick sizes (futures 0.25–0.5bp, Kalshi 0.25bp per cent); quotes, not trades.

After a large move, the other market follows only partly

Cumulative change divided by the size of the move, top 0.1% of 1-minute moves



Futures moves (210, avg. 1.29bp): Kalshi follows 16% within 15 minutes, 20% within 2 hours. Kalshi moves (195, avg. 0.58bp): Kalshi builds to 160% and futures follow 49% within 2 hours.